S. B. JAIN INSTITUTE OF TECHNOLOGY, MANAGEMENT & RESEARCH, NAGPUR.

Practical No. 6

Aim: To design and implement an Artificial Neural Network for classification problem and evaluate its learning performance.

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Aim: To design and implement an Artificial Neural Network for classification problem and evaluate its learning performance.

OBJECTIVE/EXPECTED LEARNING OUTCOME:

The objectives and expected learning outcome of this practical are:

It is designed to analyze and process information as humans. Artificial Neural Network has self-learning capabilities to produce better results as more data is available.

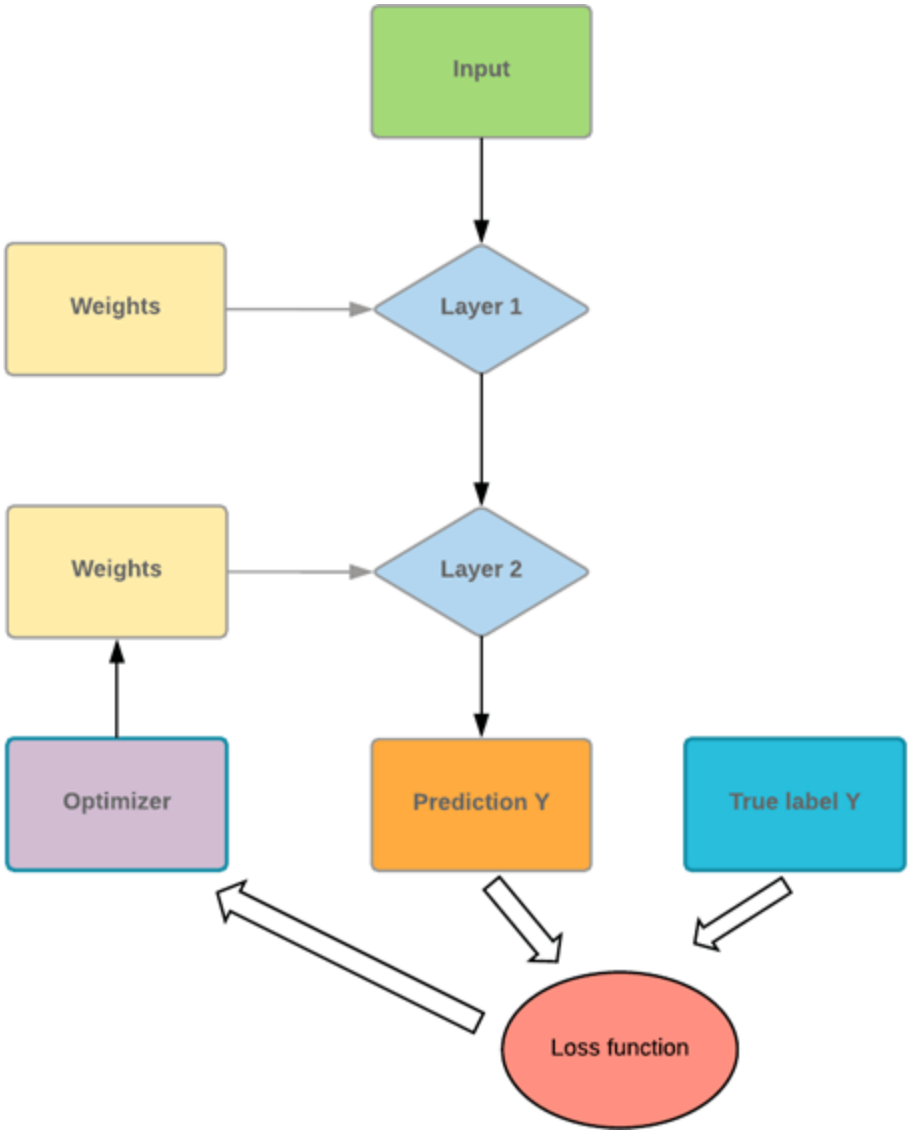
THEORY:

Neural networks are parallel computing devices, which is basically an attempt to make a computer model of the brain. The main objective is to develop a system to perform various computational tasks faster than the traditional systems. These tasks include pattern recognition and classification, approximation, optimization, and data clustering.

ANNs (Artificial Neural Network) is at the very core of Deep Learning an advanced version of Machine Learning techniques. ANNs are versatile, adaptive, and scalable, making them appropriate to tackle large datasets and highly complex Machine Learning tasks such as image classification (e.g., Google Images), speech recognition (e.g., Apple's Siri), video recommendation (e.g., YouTube), or analyzing sentiments among customers (e.g. Twitter Sentiment Analyzer).

What is an Artificial Neural Network?

An **Artificial Neural Network** (ANN) is a computer system inspired by biological neural networks for creating artificial brains based on the collection of connected units called artificial neurons. It is designed to analyse and process information as humans. Artificial Neural Network has self-learning capabilities to produce better results as more data is available.



Artificial Neural Network

An Artificial Neural Network (ANN) is composed of four principal objects:

- **Layers**: all the learning occurs in the layers. There are 3 layers 1) Input 2) Hidden and 3) Output
- **Feature and label**: Input data to the network (features) and output from the network (labels)
- **Loss function**: Metric used to estimate the performance of the learning phase
- **Optimizer**: Improve the learning by updating the knowledge in the network

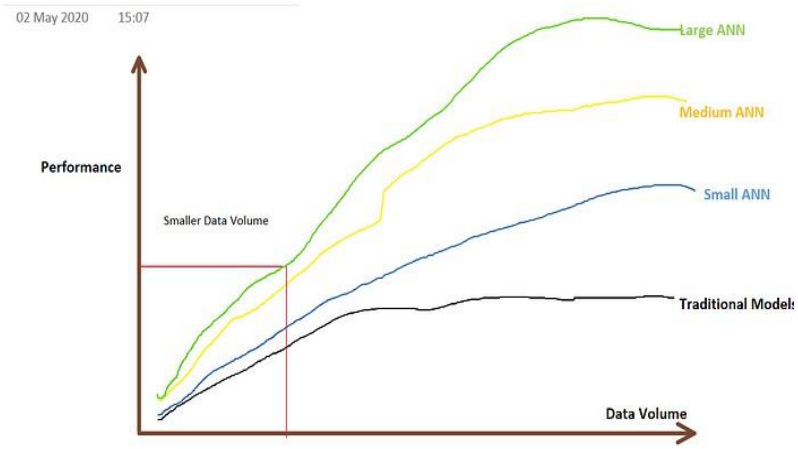
A neural network will take the input data and push them into an ensemble of layers. The network needs to evaluate its performance with a loss function. The loss function gives to the network an idea of the path it needs to take before it masters the knowledge. The network needs to improve its knowledge with the help of an optimizer.

If you take a look at the figure above, you will understand the underlying mechanism.

The program takes some input values and pushes them into two fully connected layers. Imagine you have a math problem, the first thing you do is to read the corresponding chapter to solve the problem. You apply your new knowledge to solve the problem. There is a high chance you will not score very well. It is the same for a network. The first time it sees the data and makes a prediction, it will not match perfectly with the actual data.

To improve its knowledge, the network uses an optimizer. In our analogy, an optimizer can be thought of as rereading the chapter. You gain new insights/lesson by reading again. Similarly, the network uses the optimizer, updates its knowledge, and tests its new knowledge to check how much it still needs to learn. The program will repeat this step until it makes the lowest error possible.

In our math problem analogy, it means you read the textbook chapter many times until you thoroughly understand the course content. Even after reading multiple times, if you keep making an error, it means you reached the knowledge capacity with the current material. You need to use different textbook or test different method to improve your score. For a neural network, it is the same process. If the error is far from 100%, but the curve is flat, it means with the current architecture; it cannot learn anything else. The network has to be better optimized to improve the knowledge.



Above figure illustrates the Variation of performance with an increase in data volume. Most algorithms produce similar performance measures for smaller datasets, however, Neural

Networks works best once the data volume expands beyond a certain threshold. (image developed by the author using one note)

Program for reference:

```
X = data_frame.drop(columns='label', axis=1)
Y = data_frame['label']
X_train, X_test, Y_train, Y_test = train_test_split(X, Y, test_size=0.2, random_state=2)
print(X.shape, X_train.shape, X_test.shape)
from sklearn.preprocessing import StandardScaler
scaler = StandardScaler()

X_train_std = scaler.fit_transform(X_train)
X_test_std = scaler.transform(X_test)
import tensorflow as tf
tf.random.set_seed(3)
from tensorflow import keras
model = keras.Sequential([
                    keras.layers.Flatten(input_shape=(30,)),
                    keras.layers.Dense(20, activation='relu'),
                    keras.layers.Dense(2, activation='sigmoid')
])
model.compile(optimizer='adam',
          loss='sparse_categorical_crossentropy',
          metrics=['accuracy'])
loss, accuracy = model.evaluate(X_test_std, Y_test)
print(accuracy)
Y_pred = model.predict(X_test_std)
```

PROGRAM CODE:

OUTPUT (SCREENSHOT):

CONCLUSION:

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DISCUSSION AND VIVA VOCE:

1. When would you choose an Artificial Neural Network over traditional machine learning models like Random Forest or Logistic Regression?
2. Why do neural networks require large amounts of data in real-world applications?
3. How do you detect and prevent overfitting in an ANN deployed in production?
4. What is the role of activation functions in neural networks, and why is ReLU commonly used in industry?
5. How do you decide the number of hidden layers and neurons in an ANN?
6. Why is feature scaling mandatory in neural networks?
7. What is the vanishing gradient problem, and how does it affect deep networks?
8. How do you evaluate whether an ANN model is ready for deployment in industry?
9. What are the computational challenges of using ANN in real-time systems?
10. Why are deep neural networks considered "black box" models, and how do industries handle explainability concerns?

REFERENCE

- https://en.wikipedia.org/wiki/Artificial_neural_network
- https://www.sciencedirect.com/topics/earth-and-planetary-sciences/artificial-neural-network